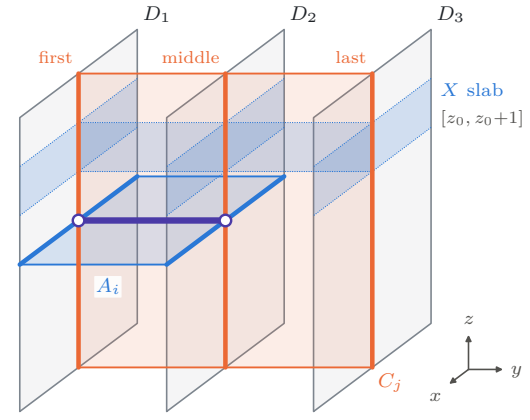
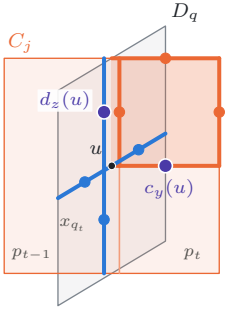
(a) Sheets, junction lines and the wall



Z junction X junction wall ℓ_{ij} switch

(b) Middle Z junction at u

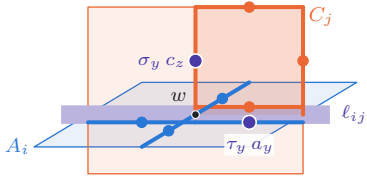


star $+ c_y(u)$, face $+ d_z(u)$

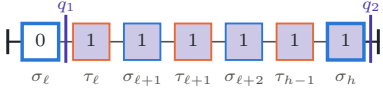
$p_{t-1} + p_t + x_{q_t} = 0$

word 110: backbone on, $x_{q_t} = 0$

(c) Wall cell on ℓ_{ij}



fixed ends, a switch at each $q \in X_i \cap Z_j$



solvable iff $|X_i \cap Z_j|$ is even